

SESSION-03
SECURITIES ANALYSIS & PORTFOLIO MANAGEMENT

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

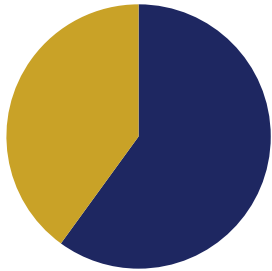


Dr. (CMA) Srikanth Potharla, M.Com., M.Phil., FCMA., UGC-NET., Ph.D.

Assistant Professor, Department of Accounting & Finance · ICAI Business School (IFHE Hyderabad)

Portfolio Return: The Weighted Average That's Easy to Compute

- $E(R_p) = \sum w_i E(R_i)$ — the capital-weighted average of individual expected returns
- Weights (w_i) reflect capital allocation — how much money is invested in each asset
- Diversification does not change expected portfolio return — combining assets never creates or destroys expected return by itself
- The “magic” of diversification, whatever it turns out to be, must therefore happen entirely on the risk side



■ Asset A (60%) ■ Asset B (40%)

Asset A: $w=60\%$, $E(R)=14\%$

Asset B: $w=40\%$, $E(R)=8\%$

PORTFOLIO EXPECTED RETURN

$$E(R_p) = 0.6(14\%) + 0.4(8\%) = 11.6\%$$

A pure weighted average — deliberately plain, because that's all it is. No covariance, no correlation, no interaction terms.

REFLECTION POINT

“If diversification doesn't raise or lower expected return, why does almost every investor intuitively believe diversification ‘reduces returns to be safe’? Where does that belief come from?”

QUESTION BANK LINK

Use in class: Problem 3.1 (portfolio return, two-asset portfolio)

Portfolio Risk: Why It Is Not a Weighted Average

- $\sigma_p^2 = \sum \sum w_i w_j \sigma_{ij}$ — portfolio variance captures every pairwise covariance interaction, not just individual variances
- Key insight: two portfolios with identical expected return can justify completely different capital allocation decisions once correlation is factored in
- For n assets: $(n^2 - n)/2$ distinct covariance pairs must be considered

PORTFOLIO 1

$$E(R_p) = 10\%$$

$$\sigma_p = 13.9\% \quad (\rho = +0.8)$$

$$\text{Sharpe ratio} = 0.43$$

PORTFOLIO 2

$$E(R_p) = 10\%$$

$$\sigma_p = 9.1\% \quad (\rho = -0.3)$$

$$\text{Sharpe ratio} = 0.66$$

Identical expected return, meaningfully different risk — and a meaningfully different Sharpe ratio.

REFLECTION POINT

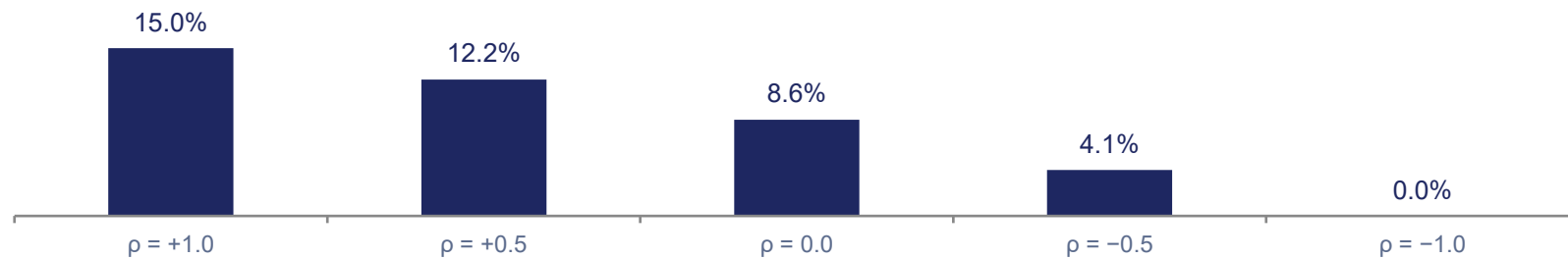
“Two portfolios show the identical 10% expected return on a fact sheet. What single number would you ask for next before deciding between them — and why isn’t that number on most retail fact sheets?”

QUESTION BANK LINK

None directly on this slide (conceptual formula slide) — numerical anchors follow immediately in Slide 25 (Problems 3.2–3.4).

Portfolio Risk Under Three Correlation Regimes

- Same two-asset portfolio, three scenarios: $\rho=+1.00$ (perfect positive) | $\rho=+0.65$ (positive, imperfect) | $\rho=-0.65$ (negative)
- Diversification benefit strengthens as correlation falls — weakest at $\rho=+1.00$, strongest at $\rho=-0.65$
- Correlation, not volatility, is often the more decision-relevant variable in portfolio construction

PORTFOLIO RISK (σ_p) ACROSS THE CORRELATION SPECTRUMQB CASES (σ_p , ILLUSTRATIVE)

$\rho = +1.00 \rightarrow 15.0\%$

$\rho = +0.65 \rightarrow \approx 11.4\%$

$\rho = -0.65 \rightarrow \approx 3.0\%$

REFLECTION POINT

"If correlation is this powerful, why do so many retail 'diversified' portfolios still hold assets that are all highly positively correlated with each other?"

QUESTION BANK LINK

Use in class: Problem 3.2 ($\rho=+1.00$, perfect positive)

Assign for practice: Problems 3.3 ($\rho=+0.65$), 3.4 ($\rho=-0.65$)

The Diversification Revolution: Markowitz's 1952 Insight

- Harry Markowitz (1952): shifted investing from security analysis to portfolio analysis
- Diversification reduces risk through covariance mathematics — not through any kind of “magic”
- Historical wealth destruction (Enron, Lehman) arose from concentration, not simply from bad fundamentals
- Central claim: most investment failures are portfolio failures, not security failures

UNSYSTEMATIC RISK — DIVERSIFIABLE

Firm-specific. Combine enough uncorrelated holdings and this component shrinks toward zero — without giving up expected return.

SYSTEMATIC RISK — NON-DIVERSIFIABLE

Market-wide. No amount of diversification touches this floor — it is the risk the market actually compensates you for bearing.

“Most investment failures are portfolio failures, not security failures.” — Enron, Lehman: concentration, not just bad fundamentals, destroyed wealth.

REFLECTION POINT

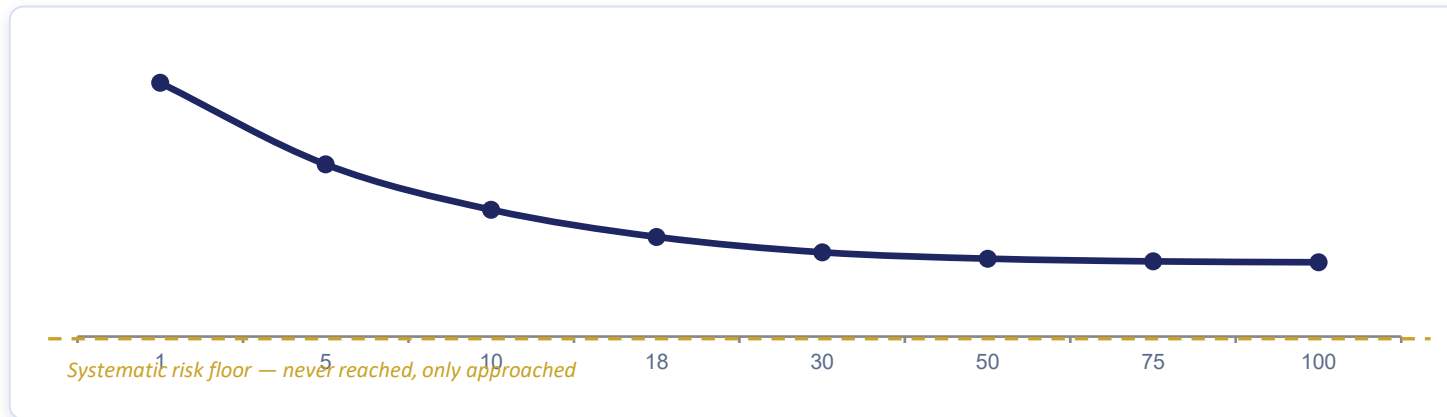
“Is diversification truly the only free lunch in finance, or does it come at the cost of conviction-based returns? Where would Warren Buffett stand on this question?”

QUESTION BANK LINK

None directly — this slide is the conceptual anchor for the Efficient Frontier problems (Section 5) in Session 4.

How Many Stocks Are Enough? Diversification Then and Now

- Traditional view (1960s): 12–18 stocks capture roughly 90% of achievable diversification benefit
- Modern evidence: rising idiosyncratic volatility now requires 50+ stocks for the same effect (Campbell, Lettau, Malkiel & Xu)
- Buffett's counter-argument: “Diversification is protection against ignorance” — concentrated, high-conviction portfolios can outperform if manager skill is genuine



12–18 stocks

~ 90% of the benefit (1960s evidence)

50+ stocks

Needed today — idiosyncratic volatility has structurally risen

REFLECTION POINT

“In the Indian context, why do retail investors often hold 2–3 stocks and describe it as ‘research-based concentration’ rather than simply undiversified risk?”

QUESTION BANK LINK

None — conceptual extension of the diversification theme.

When Diversification Fails: Correlation Spikes in a Crisis

- In a crisis, the operative question shifts from “what should I sell?” to “what CAN I sell?” — forced selling by leveraged investors drives correlations toward 1
- Even gold failed as a hedge in the first weeks of the COVID crash

2008

Global Financial Crisis

Correlation: **0.82**

Max Drawdown: **-52%**

2020

COVID Crash

Correlation: **0.79**

Max Drawdown: **-38%**

2022

Rate Shock

Correlation: **0.68**

Max Drawdown: **-24%**

the classic 60/40 portfolio failed — bonds and equities fell together

“Diversification’s promise is weakest exactly when it’s needed most.”

REFLECTION POINT

“Does diversification provide false confidence in calm markets? If historical correlation cannot be trusted in a crisis, what should investors actually do differently?”

QUESTION BANK LINK

None — real-world stress-test extension of Slide 25’s correlation-regime concept.

The Covariance Equation Behind Every Optimizer

- Individually “safe” assets can combine into a risky portfolio, and vice versa — portfolio risk is an emergent property of the entire covariance matrix, not any one asset
- Scaling problem: 100 securities require 4,950 distinct correlation pairs to be estimated
- Individual security analysis is structurally insufficient once portfolios reach institutional scale

$$\text{Cov}(R_i, R_j) = \rho_{ij} \times \sigma_i \times \sigma_j$$

correlation

volatility of i volatility of j 

REFLECTION POINT

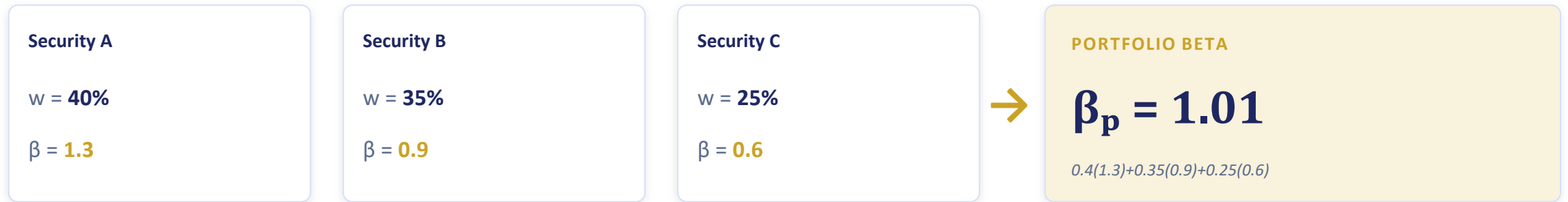
“If a 100-security portfolio needs 4,950 correlation estimates, and each one carries some estimation error, what does that imply about how much you should actually trust an ‘optimized’ portfolio built from historical data?”

QUESTION BANK LINK

None directly — this slide synthesizes Problems 3.2–3.4 and 4.3–4.4 conceptually as a unifying equation.

Portfolio Beta: Aggregating Systematic Risk Across Holdings

- Portfolio beta: $\beta_p = \sum w_i \beta_i$ — the weighted average of individual security betas
- Extends Session 2's single-security beta concept to the entire portfolio — portfolio-level betas are far more stable over time
- Fund managers use portfolio beta as a single dial for tactical exposure — “loading beta” in bull markets, “reducing beta” in bear markets



REFLECTION POINT

“If portfolio-level betas are far more stable than individual-stock betas, does that make portfolio beta a more trustworthy tactical tool, or does it just hide instability that's still there underneath at the security level?”

QUESTION BANK LINK

Use in class: Problem 9.4 or 9.5 (weighted portfolio beta)

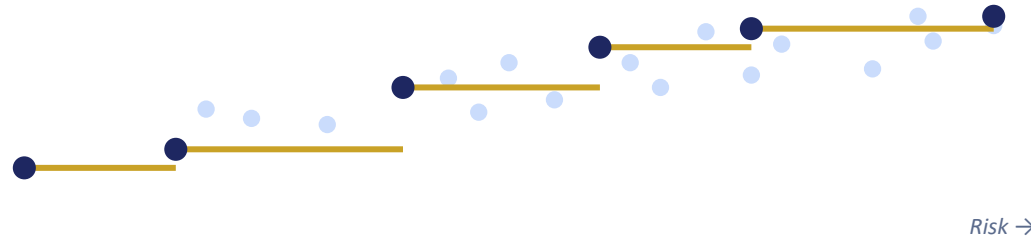
Assign for practice: Problem 9.6 (beta overlay via index futures — advanced/challenge)

Session 3 Synthesis: Setting Up the Efficient Frontier

- Portfolio return (simple, weighted average) vs portfolio risk (covariance-driven) — the session's central contrast
- Markowitz's insight: diversification reduces risk mathematically, not magically, through correlation and covariance
- Correlation regimes, crisis correlation spikes, and portfolio beta all extend this single idea in different directions

EVERY DOT = ONE FEASIBLE PORTFOLIO

Return ↑



OPEN QUESTION

Which of these feasible combinations would a rational investor actually choose?

REFLECTION POINT

"If we could plot every possible combination of these assets on a risk-return graph, which combinations would a rational investor never choose — and why?"

QUESTION BANK LINK

None — pure bridge/synthesis slide closing Session 3.